TEXORA · LECTURE SHEET

Winding Calculations

Winding looks like the simplest process in the mill — yarn in, package out — but its calculations decide how much yarn a spindle delivers per shift, how heavy a package can be built, and whether the package will survive the next process. This sheet works through surface speed, production, package build and winding density, the four calculations a weaving or knitting technologist actually uses.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 1 · SPECIALIZATION — FABRIC MANUFACTURING

1. What the calculation must answer

Winding transfers yarn from small supply packages (ring bobbins, cops, hanks) onto larger delivery packages — cones, cheeses, pirns — while clearing faults and joining breaks. Every winding calculation reduces to one of four questions: **(i)** how fast is the yarn travelling (surface speed)? **(ii)** how much yarn does one spindle produce in a shift (length and weight)? **(iii)** how much yarn can a package of given build hold? **(iv)** how tightly is it packed (winding density), which decides dyeing behaviour and unwinding performance downstream?

2. Surface speed of a drum-driven winder

On the common drum (friction) winder the package rests on a rotating drum and is turned by surface contact, so the yarn speed equals the drum's surface speed as long as there is no slip:

$$\text{Yarn speed } v \text{ (m/min)} = \pi \times D \text{ (m)} \times N \text{ (rpm)}$$

WORKED EXAMPLE — DRUM SURFACE SPEED

Drum diameter 80 mm, drum speed 3,200 rpm.

$v = \pi \times 0.080 \times 3{,}200 =$ **804 m/min**.

This is the ceiling for the winding speed of that head: the yarn can never be wound faster than the surface that drives it, and in practice slip and tension control keep the delivered speed below this figure. Precision winders, which drive the package spindle directly, are calculated the other way — from spindle rpm and the *instantaneous* package diameter, which grows as the package builds.

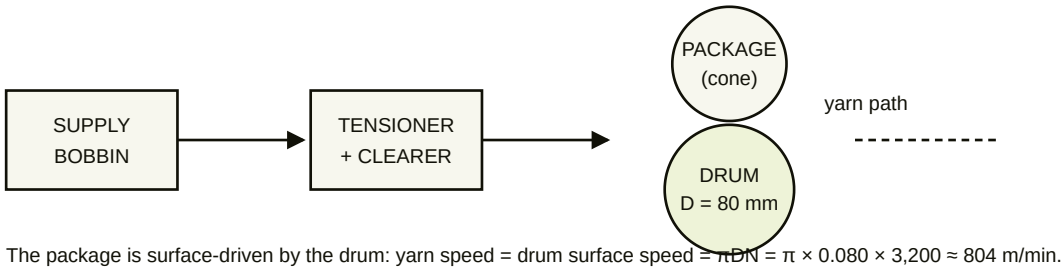


Figure 1 — Yarn path on a drum winder and the surface-speed relation of the worked example.

3. Production per spindle

Length wound in a shift is speed × time × efficiency; weight follows from the count. Efficiency here is a measured quantity — it absorbs doffing, break-mending, creeling and waiting — and on automatic winders typically lies between 75% and 90%.

$$\text{Length per shift} = v \times 60 \times \text{hours} \times \text{efficiency}$$

$$\text{Weight (kg)} = \text{length (m)} \times \text{tex} \div 1{,}000{,}000$$

WORKED EXAMPLE — PRODUCTION OF ONE WINDING HEAD

Yarn Ne 20 cotton (29.53 tex), set winding speed 1,200 m/min, efficiency 85%, 8-hour shift.
Length = 1,200 × 60 × 8 × 0.85 = **489,600 m per spindle per shift**.
Weight = 489,600 × 29.53 ÷ 1,000,000 = **14.46 kg per spindle per shift**.
A 60-head machine on this duty therefore delivers about 60 × 14.46 ≈ 868 kg per shift — the figure the planner uses in Sheet 6, where winding capacity is matched against the whole weaving shed's demand (there the same formula, at 1,100 m/min and 80% over a 22-hour day, gives 28.58 kg per spindle per day).

Reading the formula backwards is just as common: given a daily quantity and a deadline, the required number of spindles is quantity ÷ production per spindle, rounded *up* and then up again to a whole machine or section — never down.

4. Package build — how much yarn a package holds

A cylindrical cheese is an annular block of yarn: outer diameter D, core (barrel) diameter d, traverse length L. Its yarn volume, times the winding density, gives the package weight:

$$V = (\pi/4) \times (D^2 - d^2) \times L \quad ; \quad \text{Weight} = V \times \text{winding density}$$

WORKED EXAMPLE — WEIGHT OF A CHEESE PACKAGE

Cheese of outside diameter 175 mm, core 60 mm, traverse 152 mm, wound at a density of 0.42 g/cm³.

$V = (\pi/4) \times (17.5^2 - 6.0^2) \times 15.2 = (\pi/4) \times 270.25 \times 15.2 \approx$ **3,226 cm³**.

Weight = 3,226 × 0.42 ≈ **1,355 g ≈ 1.36 kg**.

At 29.53 tex this package carries 1,355 ÷ 29.53 × 1,000 ≈ 45,900 m of yarn — roughly nine warping-beam lengths of Sheet 2 will be taken from packages like this one, which is why package weight is a planning number, not a curiosity.

Winding density is set by winding tension and package hardness. Typical densities: soft dye packages 0.32–0.38 g/cm³ (liquor must flow through them), ordinary weaving cones 0.40–0.45, hard pirns higher still. The same build at 0.35 instead of 0.42 would hold only 1,129 g — density is capacity, and it is bought with tension, which is why tension is the winding technologist's main control and main fault source.

5. Traverse, crossing and patterning

The yarn is laid in a helix by the traverse. Where successive coils fall almost exactly on top of each other — when the traverse ratio (package revolutions per double traverse) is near a whole number — the package **patterns** (ribbons): hard ridges form, layers slip at doffing and the package unwinds badly. Anti-patterning devices (drum groove design, package rocking, electronic ratio stepping) exist to keep the ratio away from whole numbers through the build. On drum winders the crossing angle is fixed by the drum groove; on precision winders the traverse ratio is chosen and then stepped down as the package grows, so the wind angle falls through the build — the calculation behind "stepped precision winding".

Glossary

Surface speed — linear speed of a rotating drum's surface, πDN. **Traverse** — the axial to-and-fro motion that spreads yarn across the package. **Winding density** — package yarn mass per unit package volume, g/cm³. **Patterning (ribboning)** — coils stacking in register when the traverse ratio nears a whole number. **Traverse ratio** — package revolutions per double traverse.

Key points

• Drum winder: yarn speed = πDN (worked: 804 m/min ceiling).

• Production: length = v × time × efficiency; weight = length × tex ÷ 10^6 (worked: 489,600 m = 14.46 kg/spindle/shift).

• Package weight = annular volume × density (worked: 3,226 cm^3 × 0.42 ≈ 1.36 kg).
• Density is a process decision: dye packages soft (0.32–0.38), weaving cones 0.40–0.45 g/cm^3.

Practice

1. A cone holds 1.5 kg of Ne 30 yarn (19.68 tex). What length of yarn is on the cone?
2. A winder runs at 950 m/min, 80% efficiency, for a 22-hour day on Ne 24 (24.6 tex). Find the production per spindle per day in kg.
3. The cheese of Section 4 is rewound as a dye package at density 0.35 g/cm^3. Find its new weight and the length of Ne 20 (29.53 tex) it holds.
4. Explain why a planner who needs 42.3 spindles by calculation installs at least 48.

Answer key

1. Length = 1,500 g ÷ 19.68 g/km ≈ **76.2 km (≈ 76,200 m)**.
2. Length = 950 × 60 × 22 × 0.80 = 1,003,200 m; weight = 1,003,200 × 24.6 ÷ 10^6 = **24.68 kg**.
3. Weight = 3,226 × 0.35 = **1,129 g**; length = 1,129 ÷ 29.53 × 1,000 ≈ **38,230 m**. Softer build, less yarn — the price of dye liquor flow.
4. Spindles come in machine sections, not singles; the installed number must also absorb breakdowns, count changes and creel changeover, so the planner rounds up to the next whole machine/section and keeps a margin — a winder run at 100% of calculated capacity has no capacity left for reality.